MIL-A-15199A(SHIPS)
19 February 1952
SUPERSEDING
MIL-C-15199(SHIPS)
1 September 1950

MILITARY SPECIFICATION

ADHESIVE, SILICATE, CEMENT; INSULATION FIBROUS

(All interested bureaus of the Navy Department have concurred in the use of this specification for purchase.)

1. SCOPE

1.1 This specification covers fibrous insulation adhesive cement for securing woven asbestos cloth to insulating material employed on piping or other applications for Naval vessels.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

SS-C-466 - Cloth, Yarn, Thread, and Tape; Asbestos.
LLL-B-631 - Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL-B-636 - Boxes; Fiber, Solid (for Domestic Shipment).

MILITARY

JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed.
MIL-B-107 - Boxes, Wood, Wirebound (Overseas Type).
JAN-P-124 - Packaging and Packing for Overseas Shipment - Containers (Cans, Pails, and Drums), Metal (for Other Than Subsistence Items).
MIL-C-2861 - Cement, Insulation, High-Temperature.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-129 - Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - Fibrous adhesive cement shall be composed of a mixture of sodium silicate and short asbestos fibers thoroughly mixed to obtain uniform distribution of the ingredients. The cement shall not contain any other materials which would have a deleterious effect upon metal, asbestos lagging and thermal insulation to which applied.

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3.1.1 Cement, in the condition in which delivered, shall be suitable and effective for the purpose intended without heating or addition of other ingredients. The cement shall not deteriorate when stored for a period of one year in airtight containers.

3.2 Adhesive strength. - The adhesive strength of the cement shall be such that, when tested by the method specified in 4.4.1, a separation of the applied cloth from the insulating material will result in rupture of the insulating material and not of the cement.

3.3 When tested by the method specified in 4.4.2, the cement shall not give off objectionable or irritating fumes at temperatures up to 200° Fahrenheit (F.).

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Lot. - For purposes of sampling, a lot shall consist of all cement from one production batch offered for delivery at one time.

4.2.1 Sampling. - The Government inspector shall select a representative 1-gallon sample from each lot. The sample so selected shall be forwarded to a Government laboratory designated by the bureau or agency concerned for the tests specified in 4.4. Samples shall be carefully identified as to manufacturer's name, contract number, and lot number.

4.3 Rejection. - If the sample selected in accordance with 4.2.1 fails any of the tests specified in 4.4, the entire lot shall be rejected. A rejected lot may be resubmitted for Government inspection, but only after the manufacturer, subsequent to the first Government inspection, has reworked the entire lot; subjected the lot to the test or tests wherein failure occurred, and has thereby removed all nonconforming material.

4.4 Test procedures. -

4.4.1 Adhesive strength. - High-temperature insulation cement conforming to Specification MIL-C-2861, shall be applied between the fixed flanges on a section of pipe of the size shown on figure 1, so that the cement will be flush with the flanges. After this material has become dry, the fibrous cement under test shall be used to apply to the surface of this insulating material, three strips of grade A asbestos cloth, conforming to Specification SS-C-466. Each strip shall be 3 inches wide by 17 inches long and shall be applied to encircle the insulation material, except that the surplus length of asbestos cloth strips shall be left free instead of lapped over to make a seam. The fibrous cement shall be allowed to dry for 72 hours at room temperature. The pipe section shall then be mounted at the uncovered ends on frictionless rollers so that the pipe will be free to turn about its longitudinal axis. Load shall be applied vertically downward on the free end of each strip of cloth, successively, until the strips are separated from the insulating material.

4.4.2 A steel plate, 12 inches square and 1/8 inch thick, shall be heavily coated on one side with the test cement. After drying for 7 days at 70° F. the plate shall be placed in an oven and heated to a temperature of 200° F. for 2 hours. No objectionable or irritating fumes shall be given off.

5. PREPARATION FOR DELIVERY

5.1 Packaging (for domestic and overseas shipment). - The cement shall be furnished in 1-quart, 1-gallon multiple-friction top containers, or in 5-gallon lug-cover steel drums as specified (see 6.1).

5.1.1 One-quart and one-gallon multiple-friction top containers shall conform to type IV of Specification JAN-P-124.

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5.1.2 Five-gallon lug-cover steel drums shall conform to type VI of Specification JAN-P-124.

5.2 Packing.-

5.2.1 For domestic shipment.-

5.2.1.1 Four 1-gallon containers or twelve 1-quart containers shall be packed in nailed wood or wirebound boxes conforming to Specifications JAN-P-106 and MIL-B-107, respectively, or in a style RSC corrugated or solid fiberboard box having a minimum Mullen test of 275 pounds conforming to Specifications LLL-B-631 and LLL-B-636, respectively. No strapping will be required.

5.2.1.2 Five-gallon lug-cover steel drums will require no overpacking.

5.3 Marking.- In addition to any special marking required by the contract or order, intermediate and shipping containers shall be marked with the date of manufacture of the product, and in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of containers required (see 5.1).
(c) Whether domestic or overseas shipment is required (see 5.1 and 5.2).

6.2 Fibrous insulation adhesive cement should not be applied directly on any painted metal surface or fibrous glass lagging for the reason that the composition of this cement is chemically active on these materials.

6.3 Superseding documents.- This specification supersedes MIL-C-15199(SHIPS). The latter superseded Navy Specification 52C22.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Interest:
Bureaus - ShSY

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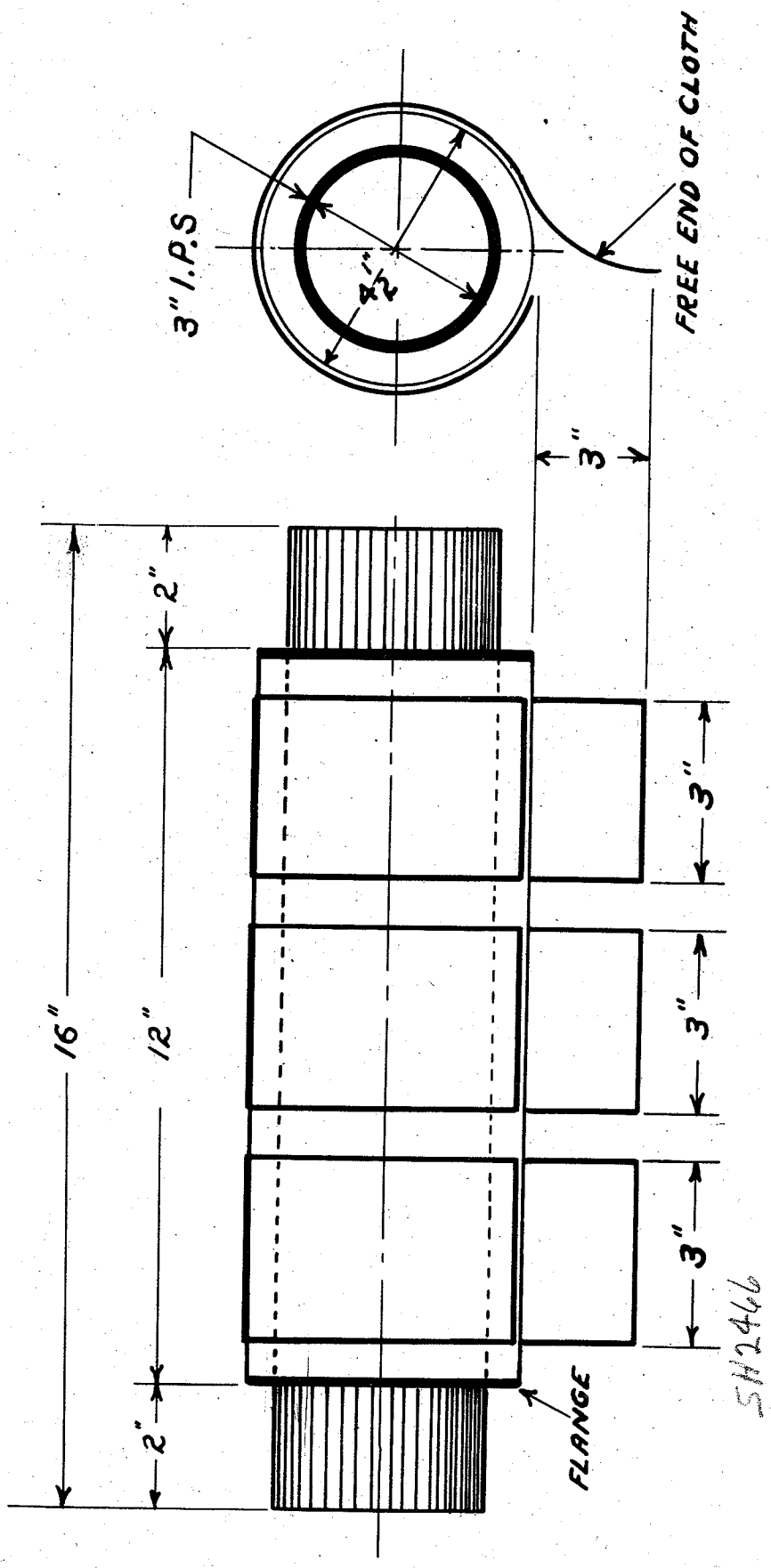


Figure 1. - Adhesive Test Apparatus.

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